EFU LAND TRANSPORT LICENCE v1.0

Experimental EFU-based Road, Truck, and Rail Transport Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for land transport (cars, trucks, rail, buses).

NOT a vehicle registration, EU CO_2 fleet regulation, or highway toll substitute, but an open research protocol measuring transport's physical footprint in human-scale EFU metrics (fuel, tire wear, noise, infrastructure).

I. Core Concepts

1. **EFU-Road (EFU-LAND)**
   EFU framework application for land transport:
   - **EFU-H_LAND**: fuel/diesel/gasoline/electricity
   - **EFU-M_LAND**: tire wear + brake dust + spare parts
   - **EFU-C_LAND**: CO_2 + black carbon (particulates)
   - **EFU-S_LAND**: noise + road/rail entropy
   - **EFU-INFRA_LAND**: road/rail maintenance (amortization)
2. **Transport Sovereignty Gap (SS_LAND)**
   SS_LAND = TEFU_LAND_total – Regional Transport Carrying Capacity (RTCC_LAND)
   Measures if transport volume stays within regional infrastructure regeneration capacity.
3. **MROI_LAND**
   MROI_LAND = THI_LAND / TEFU_LAND
   THI_LAND = economic mobility + freight transport + passenger-km efficiency.

II. EFU Land Transport Licence – Research Status

Documents EFU-audited carriers, bus companies, rail operations, fleet managers.
Does NOT replace transport permits, tachograph rules, or infrastructure fees.
Enables truck fleets, rail traffic, passenger car benchmarking.

III. Metrics and Thresholds

1. EFU Components (per passenger-km or tonne-km)

EFU-H_LAND (Fuel)
EFU-H_LAND = fuel_MWh / 12.88 / passenger_km or tonne_km
Example: Passenger car = 0.5 liter/100km → 0.00015 EFU/pax-km

EFU-M_LAND (Tires/Brakes)
EFU-M_LAND = tire_brake_kg / 30kg/person/year / tonne_km
Example: Truck = 0.2g/km → high material consumption

EFU-C_LAND (Carbon)
EFU-C_LAND = CO2e_kg / 4t/person/year / tonne_km

TEFU_LAND = 1.0×EFU-H + 1.5×EFU-M + 1.2×EFU-C + 1.0×EFU-S + 0.8×EFU-INFRA

2. Regional Transport Carrying Capacity (RTCC_LAND)

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RTCC_LAND = Sustainable_traffic_billion_tkm / Population / 10,000 tkm/person/year
Example: Germany = 500B tkm / 84M / 10,000 = 0.60 EFU_LAND/person/year
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IV. Licence Categories (EFU-LAND Labels)

Level	SS_LAND	MROI_LAND	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-LAND SOVEREIGN**	Within infra capacity
2	0 < ≤30k EFU	≥5e-3	**EFU-LAND BALANCED**	Monitorable
3	30k < ≤150k EFU	≥1e-3	**EFU-LAND DEFICIT**	Infrastructure loading
4	>150k EFU	<1e-3	**EFU-LAND CRITICAL**	Road/system limits

V. Regional RTCC_LAND Calibrations

Country	Traffic (B tkm/year)	Population (M)	RTCC_LAND (EFU/person/year)
GERMANY	500	84	0.60
USA	5,000	340	0.15
CHINA	6,000	1,410	0.04
JAPAN	400	125	0.26
EU	2,500	448	0.56

VI. Example: German Truck Traffic (DB Schenker)

Input (annual fleet):

- Tonne-km: 20B tkm
- Fuel: 8B liters (28M MWh)
- Tires/brakes: 200,000 tonnes

- Drivers: 50,000 staff
- Region: Germany (RTCC_LAND = 0.60 EFU/person/year)

Calculation:

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EFU-H_LAND = 28M/12.88/20B = 0.00011 EFU/tkm
EFU-M_LAND = 200k/30/20B = 0.000033 EFU/tkm
EFU-C_LAND = 20M tCO2/4/20B = 0.000025 EFU/tkm

TEFU_LAND_total ≈ 3,800,000 EFU/year
Allocated = 50k × 0.60 = 30,000 EFU/year
SS_LAND = 3.8M – 30k = +3,770,000 EFU/year (Level 4: CRITICAL)

Result: EFU-LAND CRITICAL
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VII. Implementation Notes

1. **EU CO_2 fleet regulation**: TEFU_LAND basis for weighted emissions
2. **Highway tolls**: EFU-M_LAND metric for tire/road wear-based pricing
3. **Data sources**: Tier 1 = fuel cards, tachographs; Tier 2 = transport statistics